

Algebraic Combinatorics Assignments

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Walks in Graphs

Problem 1 Let N_l denote the number of closed walks of length l in K_p from some vertex to itself. Notice that if we start at a random vertex v_i , then we get the following relationship between N_l and N_{l-1}

$$N_l = (p-1)^{l-1} - N_{l-1}$$

Since the $l-1$ st vertex on the walk must be a vertex other v_i . We proceed by induction on l . For $l=1,2$ we have $N_l = 0, p-1$ and the given formula agrees with these values. Assume N_{l-1} is as given in the problem statement. Then, using the relation we see that

$$N_l = (p-1)^{l-1} - N_{l-1} \iff N_l + N_{l-1} = (p-1)^{l-1}$$

Notice that

$$\frac{1}{p} \cdot ((p-1)^l + (p-1)(-1)^l + (p-1)^{l-1} + (p-1)(-1)^{l-1}) = \frac{1}{p} \cdot ((p-1)^l + (p-1)^{l-1}) = (p-1)^l$$

as desired. Thus,

$$N_l = \frac{1}{p} \cdot ((p-1)^l + (p-1)(-1)^l)$$

Problem 2 We can deduce that the eigenvalues of the adjacency matrix $A = A(G)$ are $8, 3, 3, -1, -1, -1, -6, 1, 1, 1, 1, 1, 0, 0, 0$ from Corollary 1.3 and Lemma 1.7. If we add a loop at each vertex, then the relation between the $A' = A(G')$ and A is $A' = A + I$. Notice that if v is an eigenvector of A corresponding to eigenvalue λ , then v is an eigenvector of A' with corresponding eigenvalue $\lambda + 1$. Thus, the eigenvalues of A' are $9, 4, 4, 0, 0, 0, -5, 2, 2, 2, 2, 2, 1, 1, 1$. Therefore the number of closed walks of length l in G is

$$9^l + 2 \cdot 4^l + (-5)^l + 5 \cdot 2^l + 3$$

Problem 3 Notice that in a bipartite graph, a closed walk of length l is not possible when l is odd because, starting from a vertex in A , a walk of odd length will end in a vertex in B . Let $\alpha_1, \dots, \alpha_r$ denote the positive eigenvalues of $A(G)$ and $-\beta_1, \dots, -\beta_s$ denote the negative eigenvalues of $A(G)$. We have, for odd l ,

$$\alpha_1^l + \dots + \alpha_r^l - \beta_1^l - \dots - \beta_s^l = 0$$

Rearranging the equation, we get

$$\alpha_1^l + \dots + \alpha_r^l = \beta_1^l + \dots + \beta_s^l$$

for all odd numbers l . By a similar argument in the proof of Lemma 1.7, we can conclude that $r = s$ and β_i are a permutation of the α_i s. Therefore, we see that the non-zero eigenvalues of $A(G)$ come in pairs $\pm\alpha_i$

Problem 4 The number of closed walks of length l from one of the u_i 's to itself is $s \cdot r \dots s \cdot 1 = s^{l/2} \cdot r^{l/2-1}$. Similarly, the number of closed walks of length l from one of the v_i s to itself is $r^{l/2} \cdot s^{l/2-1}$. Thus, the total number of closed walks of length l is $r \cdot r^{l/2-1} s^{l/2} + s \cdot s^{l/2-1} r^{l/2} = 2r^{l/2} s^{l/2}$. From this we can find the eigenvalues of $A(G)$.

$$\lambda_1^l + \dots + \lambda_{r+s}^l = 2r^{l/2} s^{l/2}$$

Then we can see that $\lambda_1 = \dots = \lambda_r = \sqrt{s} r^{1/2-1/l}$ and $\lambda_{r+1} = \dots = \lambda_{r+s} = \sqrt{r} s^{1/2-1/l}$

Problem 5 The adjacency graph of H_{nn} looks like

$$\begin{bmatrix} 0 & A(K_n) \\ A(K_n) & 0 \end{bmatrix}$$

Let A denote $A(K_n)$. We compute the characteristic polynomial.

$$\begin{aligned} \det \begin{bmatrix} -\lambda I & A \\ A & -\lambda I \end{bmatrix} &= (-\lambda)^n \det \left(\frac{1}{\lambda} A^2 - \lambda I \right) \\ &= (-1)^n \lambda^n \det \left(\frac{1}{\lambda} I \cdot (A^2 - \lambda^2 I) \right) \\ &= (-1)^n \lambda^n \frac{1}{\lambda^n} \det(A - \lambda I) \det(A + \lambda I) \end{aligned}$$

First, notice that the result is a polynomial of degree $2n$. Moreover, roots of $\det(A - \lambda I)$ are the eigenvalues of A whereas the roots of $\det(A + \lambda I)$ are the opposite signs of the eigenvalues of A . By proposition 1.5, we know that the eigenvalues of $A = A(K_n)$ are -1 (with multiplicity $n - 1$) and $n - 1$ (with multiplicity 1). Then, we conclude that the eigenvalues of $A(H_{nn})$ are ± 1 with multiplicity $n - 1$ and $\pm(n - 1)$ with multiplicity 1 as desired.

Problem 6 Let N_l denote the number of sequences $V_{i_0}, \dots, V_{i_l}$ for which there

exists a closed walk with vertices $v_0, v_1, \dots, v_l = v_0$ (in that order) such that $v_j \in V_{i_j}$. Then, we have the following recurrence relation

$$N_l = p(p-1)^{l-1} - N_{l-1}$$

We can then deduce the open formula $N_l = p \cdot \frac{(p-1)^l - 1}{p-2}$. Since at each step, the number of possible vertices is n , we conclude that the total number of closed walks of length l in $K(n, p)$ is

$$n^l \cdot p \cdot \frac{(p-1)^l - 1}{p-2}$$

Problem 7 The adjacency matrix of $G(n)$ looks like an enlarged version of that of G where each entry G_{ij} turns into an $n \times n$ block of 1s if v_i, v_j are connected in the graph G or 0s if they are not connected. Thus, if $v = (x_1, \dots, x_p)$ is an eigenvector with corresponding eigenvalue λ , the eigenvector for the new matrix will be $v' = (x_1, \dots, x_1, x_2, \dots, x_2, \dots, x_p, \dots, x_p)$ where each x_i is repeated n times. Then, the eigenvalue corresponding to this eigenvector will be $n\lambda$. So the eigenvalues of the new matrix is $n\lambda_1, \dots, n\lambda_p$.

Problem 8 Notice that if A is the adjacency matrix of G , then

$$A(G') = \begin{bmatrix} A & I \\ I & 0 \end{bmatrix}$$

Let $u = (x_1 \dots x_p)$ be an eigenvector of A with eigenvalue λ . We want to extend u to an eigenvector of $A(G')$. Let $v = (x_1 \dots x_p, y_1 \dots y_p)$ be an eigenvector of $A(G')$ with eigenvalue λ' , then

$$A(G')v = \begin{pmatrix} \lambda x_1 + y_1 \\ \vdots \\ \lambda x_p + y_p \\ x_1 \\ \vdots \\ x_p \end{pmatrix} = \begin{pmatrix} \lambda' x_1 \\ \vdots \\ \lambda' x_p \\ \lambda' y_1 \\ \vdots \\ \lambda' y_p \end{pmatrix}$$

We need to solve the following system for λ'

$$\begin{aligned} x_i &= \lambda' y_i \\ \lambda' x_i &= \lambda x_i + y_i \end{aligned}$$

We conclude that

$$\begin{aligned} (\lambda')^2 &= \lambda \lambda' + 1 \\ \lambda' &= \frac{\lambda \pm \sqrt{\lambda^2 + 4}}{2} \end{aligned}$$

as desired.

Cubes and the Radon Transform

Problem 1

a) In the context of n -cubes, the problem is asking us to find the probability of completing a closed walk of length l starting from the vertex $(0, 0, \dots, 0)$. Then, the answer is

$$\frac{\# \text{ closed walks of length } l \text{ from } (0, \dots, 0) \text{ to itself}}{\# \text{ walks of length } l \text{ starting from } (0, \dots, 0)} = \frac{1}{n^l} \left(\frac{1}{2^n} \sum_{i=0}^n \binom{n}{i} (n-2i)^l \right)$$

b) This time we are interested in the number of walks of length l from vertex $(0, \dots, 0)$ to $(1, \dots, 1)$. Then, the answer is

$$\begin{aligned} \frac{1}{n^l} \left(\sum_{i=0}^n \sum_{j=0}^n (-1)^j \binom{n}{j} (n-2i)^l \right) &= \frac{1}{n^l} \left(\sum_{i=0}^n (n-2i)^l \left(\sum_{j=0}^n (-1)^j \binom{n}{j} \right) \right) \\ &= \frac{1}{n^l} \left(\sum_{i=0}^n (n-2i)^l (2^n - 2^{n-1}) \right) \\ &= \frac{2^{n-1}}{n^l} \left(\sum_{i=0}^n (n-2i)^l \right) \end{aligned}$$

Problem 4 Notice that $A(G)$ has 1 in its uv -entry whenever $(A^2)_{uv}$ is non-zero, which happens only when $w(u, v) = 2$ or $w(u, u) = 0$. This observation shows that

$$A(G) = \frac{1}{A_{uv}^2} (A^2 - n^2 I)$$

Then, it follows that the eigenvalues of $A(G)$ are

$$\frac{1}{A_{uv}^2} ((n - 2w(u))^2 - n^2)$$

where u, v is any two vertices with $w(u, v) = 2$.

Random Walks

Problem 2

a) The requirements imply that $A^l = kJ$ for some positive number k , where J is the matrix of 1s. The eigenvalues for kJ are 0 with multiplicity $p - 1$ and kp with multiplicity 1. Then the eigenvalues for A must be 0 with multiplicity $p - 1$ and $(kp)^{1/l}$ with multiplicity 1. This means dimension of the column space is 1. So each column is some multiple of the first column. Similarly each row is some multiple of the first row. Moreover, since $[1 \dots 1]$ is an eigenvector for kJ , it is also an eigenvector for A^l and hence for A . Also, since A is symmetric, we can write

$$A = \begin{bmatrix} a & ak_1 & ak_2 & \dots & ak_{p-1} \\ ak_1 & ak_1^2 & ak_2k_1 & \dots & ak_{p-1}k_1 \\ \vdots & & & & \\ ak_{p-1} & ak_1k_{p-1} & ak_2k_{p-1} & \dots & ak_{p-1}^2 \end{bmatrix}$$

Applying A to the eigenvector $[1 \dots 1]$ and comparing the first and second entries yields

$$a(1 + k_1 + \dots + k_{p-1}) = ak_1(1 + k_1 + \dots + k_{p-1}) \implies k_1 = 1$$

Similarly, we see that $k_i = 1$ for all $i = 1, \dots, p - 1$. This shows that the adjacency matrix is a multiple of J . Thus, the number of edges between any two vertices is the same.

b) The conditions in the problem statement imply that $M^l = \frac{1}{p}J$ where J is the $p \times p$ matrix of all 1 entries. Then this is the same situation as part a. Then, the entries of M are all equal to each other. Since each row sum up to 1, each entry must be $\frac{1}{k}$ for some positive number k , which shows the number of edges between any two vertices is the same.

Problem 4 The probability matrix corresponding to this graph is $pI + \frac{1-p}{n}A(C_n)$. Then, the eigenvalues are $E_u = p + \frac{1-p}{n}(n - 2w(u))$. Plugging in this value to formula 2.10 we get

$$(M)_{00}^l = \frac{1}{2^n} \sum_{i=0}^n \binom{n}{i} \left(p + \frac{1-p}{n}(n - 2i) \right)^l$$

Problem 5 The probability is $\frac{1}{n-1}$. Let p_k be the probability of reaching vertex v_k last. By symmetry, we see that $p_1 = p_{n-1}$. Now, for some random vertex v_k , $k \neq 0, 1$, at some point the walk will get to either the vertex v_{k-1} or v_{k+1} . Without loss of generality, let's assume that the walk is at v_{k-1} and has not visited v_{k+1} yet. Now, in order for v_k to be the last vertex to be visited, vertex v_{k+1} has to be visited before v_k . This corresponds to completing at least one travel in the reverse direction to v_{k+1} . (ie walking this way: $v_{k-1}, v_{k-2}, \dots, 0, \dots, v_{k+2}, v_{k+1}$) Then, it follows that each vertex visited so far has to be visited again. This problem is equivalent to the probability of reaching v_k last when starting from

the vertex v_{k-1} , which has the same probability as p_1 because we can simply rename the vertices. This reasoning shows that probability is evenly distributed among the $n - 1$ vertices. Therefore, $p_k = \frac{1}{n-1}$ for all $k \neq 0$.

Problem 8 The hitting time $H(u, v)$ does not depend on the vertices u and v in K_n . So let H denote $H(u, v)$ for any two vertices $u \neq v$. We can either directly go vertex v with $\frac{1}{n-1}$ probability, or we can go another vertex w with probability $\frac{1}{n-1}$ and in H number of steps reach vertex v from w . This corresponds to the following equation.

$$H = \frac{1}{n-1} \cdot 1 + (n-2) \cdot \frac{1}{n-1} \cdot H$$

Solving the above equation gives $H = n - 1$ as desired.

Problem 10 Let A denote $H(u, v)$ when $u, v \in A_1$. Let B denote $H(u, v)$ when $u \in A_1, v \in B_2$. Let C denote $H(u, v)$ when $u \in A_2, v \in A_1$. Let D denote $H(u, v)$ when $u, v \in A_2$. Then we have the following relations:

$$\begin{aligned} B &= \frac{1}{n} \cdot 1 + \frac{n-1}{n} D \\ D &= 1 + B \\ C &= \frac{1}{m} \cdot 1 + \frac{m-1}{m} A \\ A &= 1 + C \end{aligned}$$

Solving these equations give

$$B = n, D = n + 1, C = m, A = m + 1$$

Sperner's Property

Problem 2

Notice that for each chain $x_0 < x_1 < \dots < x_n$ in A , $f(x_0) < \dots < f(x_n)$ is another chain in A . Thus, f sends chains to chains. Therefore, f^{-1} must also send chains to chains. If $x < y$, then there is a chain that contains x and y and so $f^{-1}(x)$ and $f^{-1}(y)$ are part of a chain in A . If $f^{-1}(y) \leq f^{-1}(x)$ then by the order-preserving property of f , we get $y \leq x$. Since this contradicts with the assumption, we must have $f^{-1}(x) < f^{-1}(y)$, proving f^{-1} is order-preserving as well.

Problem 3

$$\begin{aligned} F(q) &= a_n q^n + \dots + a_1 q + a_0 \\ G(q) &= b_m q^m + \dots + b_1 q + b_0 \end{aligned}$$

Then,

$$F(q)G(q) = \sum_{i=0}^{n+m} c_i q^i$$

where

$$c_i = \sum_{k=0}^i a_k b_{i-k} q^i$$

Since F, G are symmetric, we see that

$$\begin{aligned} c_i &= \sum_{k=0}^i a_k b_{i-k} q^i \\ &= \sum_{k=0}^i a_{n-k} b_{m+k-i} \\ &= \sum_{l=m-i}^{m+n-i} a_{m+n-i-l} b_l \\ &= c_{m+n-i} \end{aligned}$$

which proves $F(q)G(q)$ is symmetric. For $j < \frac{m+n}{2}$, by the uni-modality of F and G , we have

$$\begin{aligned} c_{j-1} &= a_0 b_{j-1} + a_1 b_{j-2} + \dots + a_{j/2} b_{j/2} + \dots + a_{j-2} b_1 + a_{j-1} b_0 \\ &\leq a_0 b_j + a_1 b_{j-1} + \dots + a_{j/2} b_{j/2+1} + \dots + a_{j-1} b_1 + a_j b_0 \\ &= c_j \end{aligned}$$

Similar reasoning shows $c_j > c_{j-1}$ for $j > \frac{m+n}{2}$. Hence, we see that $F(q)G(q)$ is also unimodal.

Problem 6 Since P is rank-symmetric and rank-unimodal, we have

$$p_0 \leq \dots \leq p_{n/2} \geq \dots \geq p_n$$

We're assuming n is even, but the odd case follows from a very similar argument. Then, the symmetric chain decomposition has exactly $p_{n/2}$ chains. Let's name them $C_1, \dots, C_{n/2}$. An antichain can intersect any C_i only at a single point. Thus, the largest antichain has cardinality p_j . If each chain has at least 3 elements, this implies that there are at least $p_{n/2}$ many vertices of rank $p_{n/2-1}$. If $p_{n/2} = p_{n/2-1}$ then we can again find another antichain with $p_{n/2} = p_{n/2-1}$ elements. If $p_{n/2} > p_{n/2-1}$, then the second largest antichain can have at most $p_{n/2-1}$ elements. Similarly, if each of the $p_{n/2-1}$ chains have at least 5 elements, this would imply that $p_{n/2-2} = p_{n/2-1}$. Otherwise, we have $p_{n/2-2} < p_{n/2-1}$. Nevertheless, the third largest antichain can have at most $p_{n/2-2}$ elements. Similar reasoning shows that the j -th largest antichain can have at most $p_{n/2-j+1}$ elements. Thus, the largest size of the union of j antichains is equal to the largest size of a union of j antichains.

Problem 7 Let H denote the Hasse diagram of P . Let $x_0 < x_1 < \dots < x_{m-1}$ be the largest chain with m elements. Then we must have $\rho(x_i) = i$ for all $i = 0, \dots, m-1$. Then we have $P = P_0 \sqcup \dots \sqcup P_{m-1}$ as a union of m antichains.

Group actions on Boolean Algebras

1)

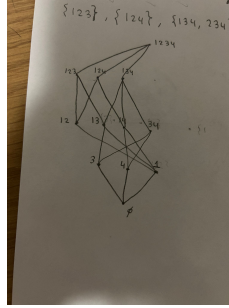


Figure 1: Hasse diagram of the quotient poset B_4/G

2)

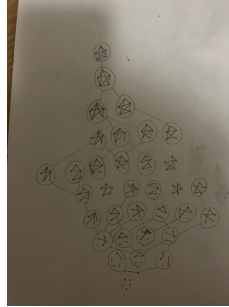


Figure 2: Hasse diagram of the poset of nonisomorphic graphs with five vertices.

3) Counterexample: Take the following poset and let $G = (67) \in S_7$. The following poset is P/G but is not Sperner since the largest antichain has only 2 elements, rather than 3.

Problem 4

Problem 5 Define another partial order on N_n by letting $u \leq' v$ if we can obtain v from u by changing some 1's to 0's. Then we see that

$$u \leq v' \iff v \leq' u$$

So, v has rank i in $(N_n, \leq)$ if and only if v has rank $n - i$ in $(N_n, \leq')$. Since this new order defined on N_n is symmetric to the order given in the problem

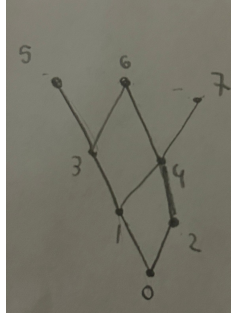


Figure 3: Counterexample.

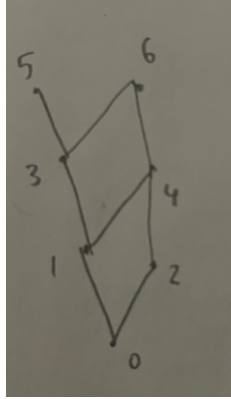


Figure 4: Hasse diagram of P/G

statement, we know that the number of elements of order i in $(N, \leq)$ is the same as those of order i in $(N, \leq')$. (There is a bijective correspondence between $P_i(N, \leq)$ and $P_i(N, \leq')$ given by $u_1 \dots u_n \mapsto (1 - u_1) \dots (1 - u_n)$). Hence, by this symmetry, we conclude that N_n is rank-symmetric, rank-unimodal, and Sperner.